

AN13989

UWB radar fundamentals

Rev. 1.2 — 30 March 2026

Application note

Document information

Information	Content
Keywords	UWB, radar, SR2XX, processing algorithm, clutter, binning, CIR, tap
Abstract	This document is explains the basics of the processing algorithm for UWB radar.



1 Introduction

1.1 Pulse radar system

In a pulse radar system, the radar transmitter sends a pulse modulating a carrier frequency. This pulse is reflected by objects in the environment and comes back to the radar receiver. The time difference between the pulse transmission and received pulse allows the radar system to determine how far the reflecting object is.

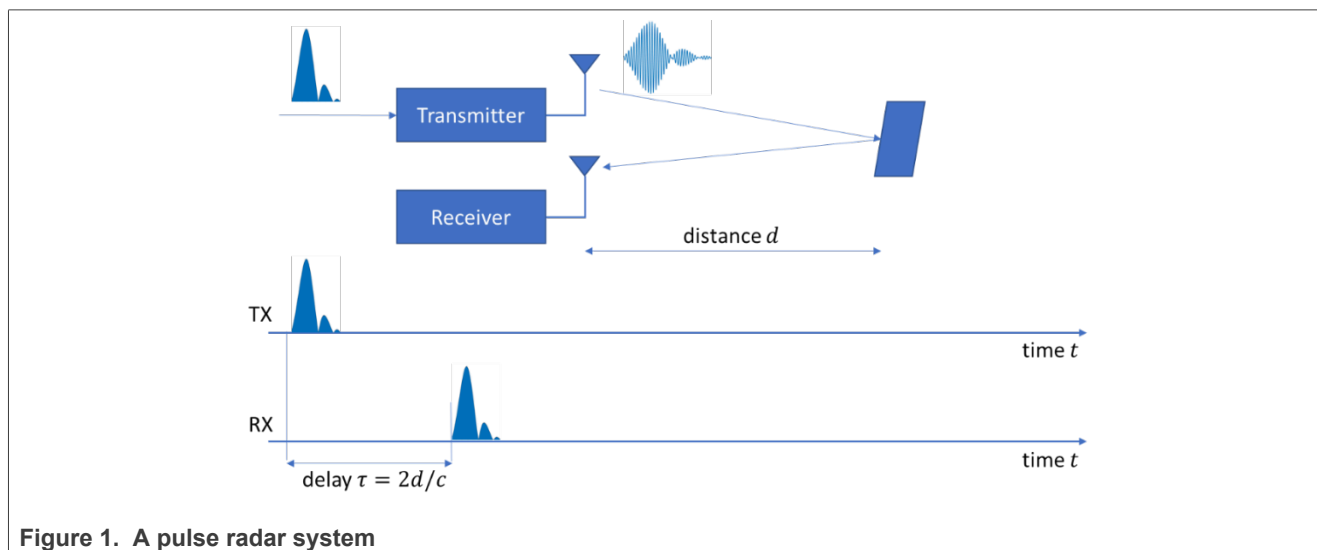


Figure 1. A pulse radar system

1.2 UWB radar sensitivity

The ability of the UWB radar system to detect a target is determined by the radar link budget:

- **Ranging use case:** TX power and RX sensitivity of both counterparts are the main contributors of the link budget. The signal attenuation in free air is also part of the equation.
- **Radar use case:** TX power and RX sensitivity of the same device are involved. The target ability to reflect the signal is also a key contributor. As the wave is traveling back and forth, the free air attenuation is contributing twice.

In the radar system:

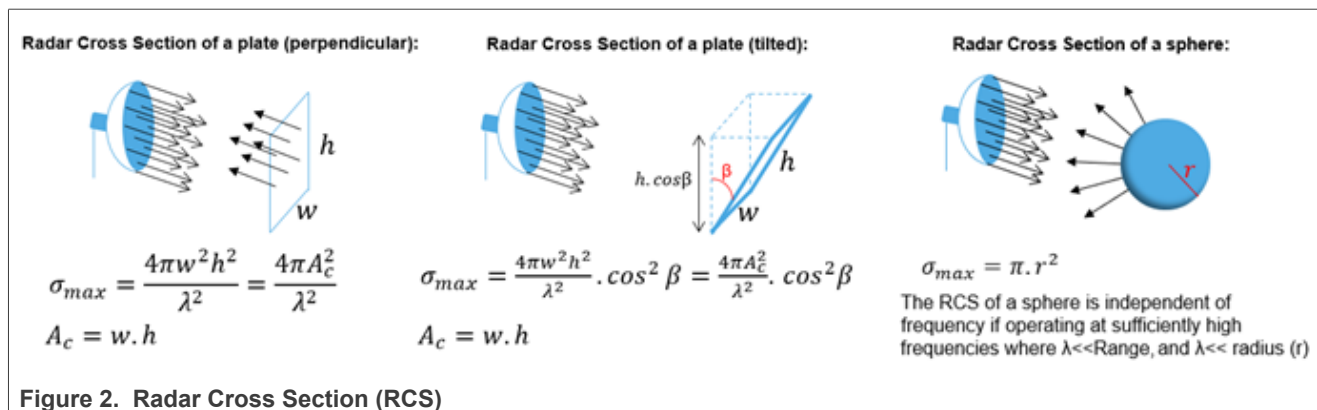
- TX power signal level is limited to RF regulation.
- Target reflection factor is governed by the target nature and dimension (Target radar Cross Section or RCS).
- Signal attenuation while traveling in free air is determined by the laws of physics.

RX sensitivity is then the only parameter managed by the device/IC.

1.3 Radar Cross Section (RCS)

Only a portion of the pulse power is reflected on the target surface. It is defined as the Radar Cross Section (RCS). The RCS of a target depends on its physical geometry, the material electrical property (reflectivity), the signal wavelength, and the incident direction of the radar beam.

The smaller the target RCS is, the lower the reflected power is and in consequence, the shorter the detection distance is.



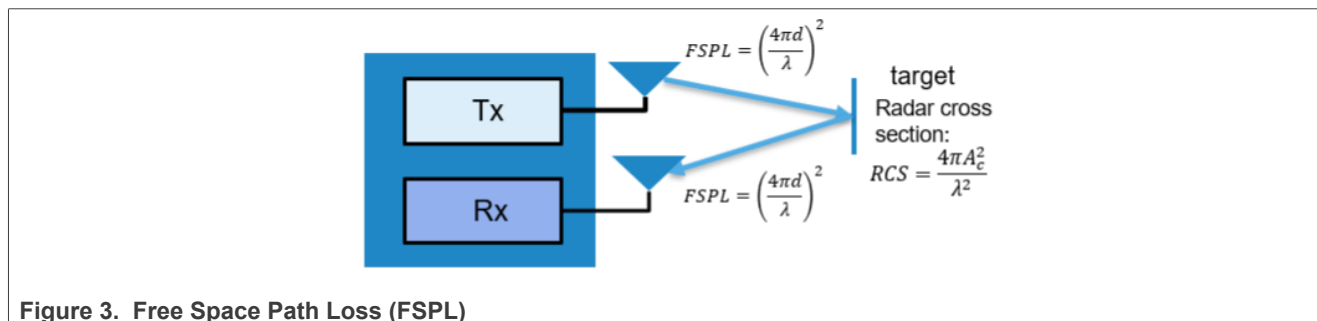
Some reference numbers:

- For breathing detection, of a static human we can consider an RCS of -16.4 dBsm (chest/lungs surface of 0.0025 m²) (dBsm refer to dB square meters).
- In case of movement detection, the RCS is 1 m² (from body movements) which is 0 dBsm.

1.4 Free Space Path Loss (FSPL)

Free Space Path Loss (FSPL) is the metric used to evaluate the attenuation of the pulse power during its propagation in free air.

As mentioned earlier, in radar, the pulse travels back and forth (two times the distance), in terms of link budget, the radar pulse is attenuated two times more than in a UWB ranging scenario.



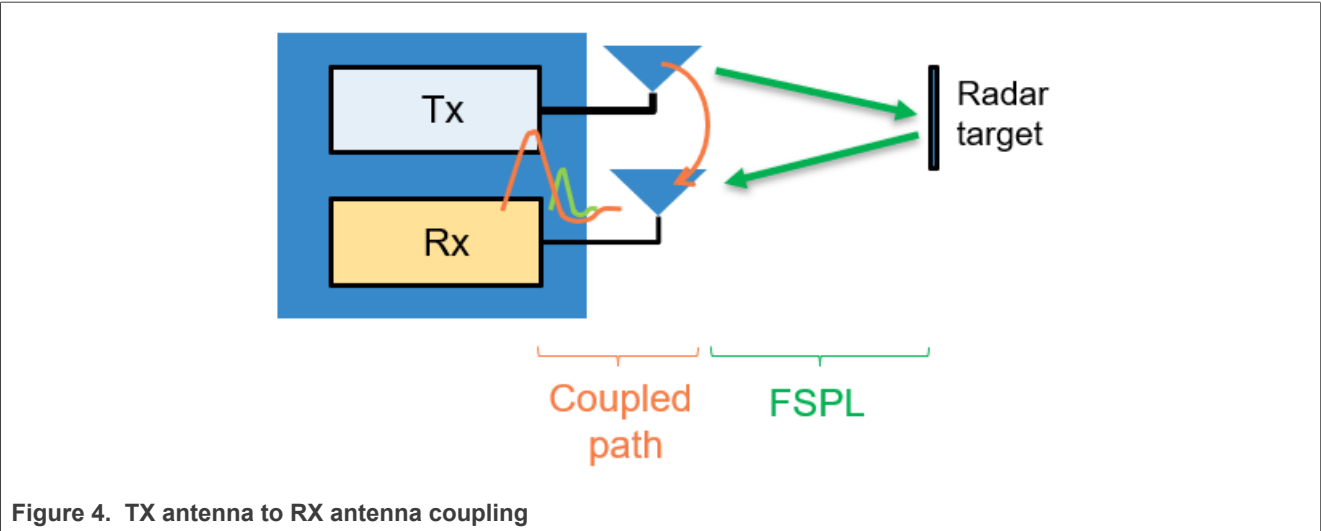
FSPL and RCS are combined into a unique formula:

$$\frac{\left(\frac{4\pi d}{\lambda}\right)^2 \times \left(\frac{4\pi d}{\lambda}\right)^2}{\left(\frac{4\pi A_c^2}{\lambda^2}\right)} = \frac{64 \times \pi^3 \times d^4}{\lambda^2 \times A_c^2}$$

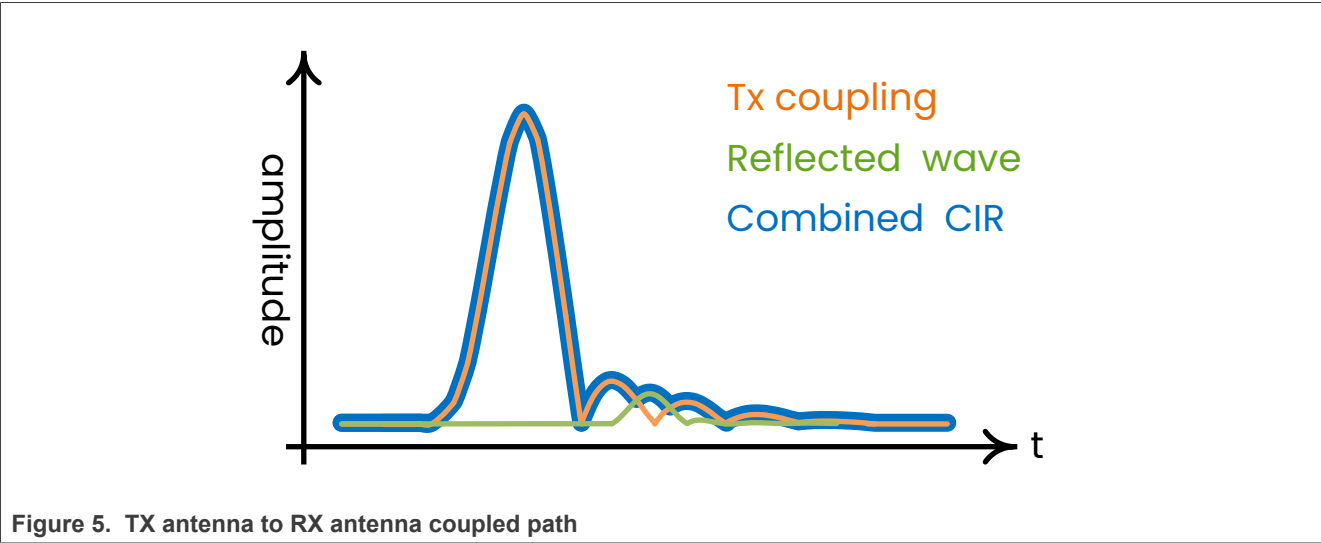
For reference, the total attenuation at 1m of a moving target (1m²) is -61.5 dB for CH9.

1.5 TX-RX coupled path

As mentioned in [Section 1.2](#), RX sensitivity is the only parameter that is not determined by regulation or physic law or target form factor and material. But RX gain could not be freely chosen. UWB radar is a full duplex system, meaning TX and RX stages are running in parallel. As a result, TX could and will pollute RX stages at different levels (chip internal coupling, board coupling, and TX/RX antenna coupling). When designing a system, keep in mind that the isolation between RX and TX is the critical parameter for an optimum RX sensitivity.



Even at a distance of 1 m, path loss for a reflected signal is approximately 60 dB. Considering 40 dB TX-to-RX antenna isolation, the coupled path is already **20 dB higher** than the reflected signal. This means the transmit power at the TX antenna is significantly higher than the received power from the target at the RX antenna. Therefore, the gain of the receiver cannot be set at its maximum, as overall power is dominated by the amount of TX power that leaks to the receiver chain. This leakage signal is referred as the “coupled path”.



2 Radar CIR (Channel Impulse Response)

The following sections are related to the Channel Impulse Response (CIR) description and interpretation.

CIR can be seen as an image of the pulse transmitted and reflected from a target. It does not consist of a single pulse but a combination ("envelop") of a train of pulses as it will be described later.

2.1 CIR tap

CIR is one measurement of the channel at a given time t .

- Radar CIR are complex numbers, from which amplitude and phase can be calculated.
- The reflected signal (pulse train) is captured and sampled at a sampling frequency of 1GHz (1 ns). 1 ns represents the resolution of the system, also referred to as "tap".
- Pulses travel at the speed of light (3×10^8 m/s). Meaning the distance covered during 1 ns is more than 30 cm. It corresponds to a distance between the radar and the reflector of 15 cm.

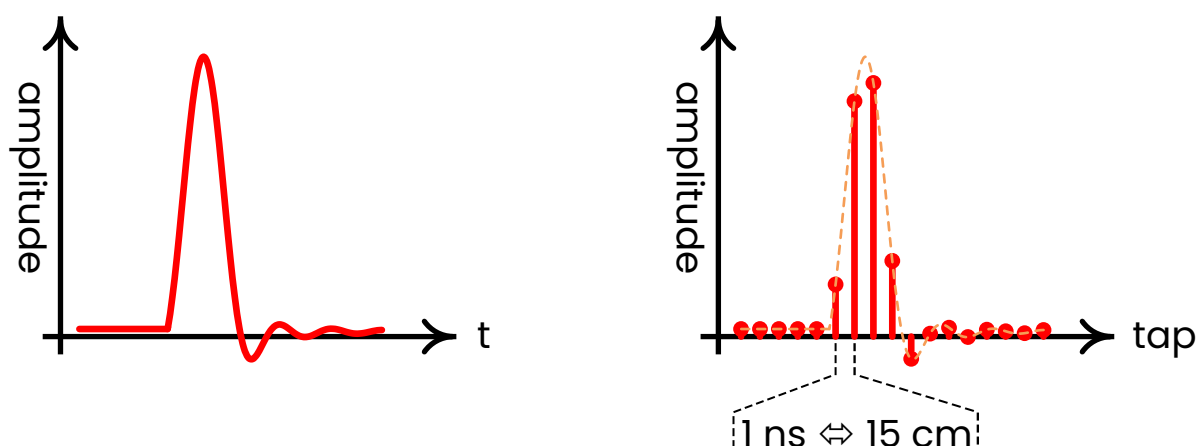


Figure 6. CIR window

2.2 CIR capture window

Considering HPRF mode of operation (refer to [Section 3.2](#)) CIR is the deconvoluted sum of the N_{sync} repetitions of a symbol composed of 91 ternary code (2ns) with a spreading factor of 4. Considering a sampling period of 1 ns, the sampled CIR length is then potentially $2 \text{ ns} \times 4 \times 91 / 1 \text{ ns} = 728$ taps. These 728 taps are stored in the SRAM.

A reduced window ("CIR capture window") is created to only keep the taps of interest.

In the NXP UWB system, only 128 taps are kept. The first eight taps are file header. The following 7 taps are samples collected before sending/receiving the radar pulse. Therefore, only 113 taps are relevant for the CIR (which corresponds to a max radar distance of 17 m).

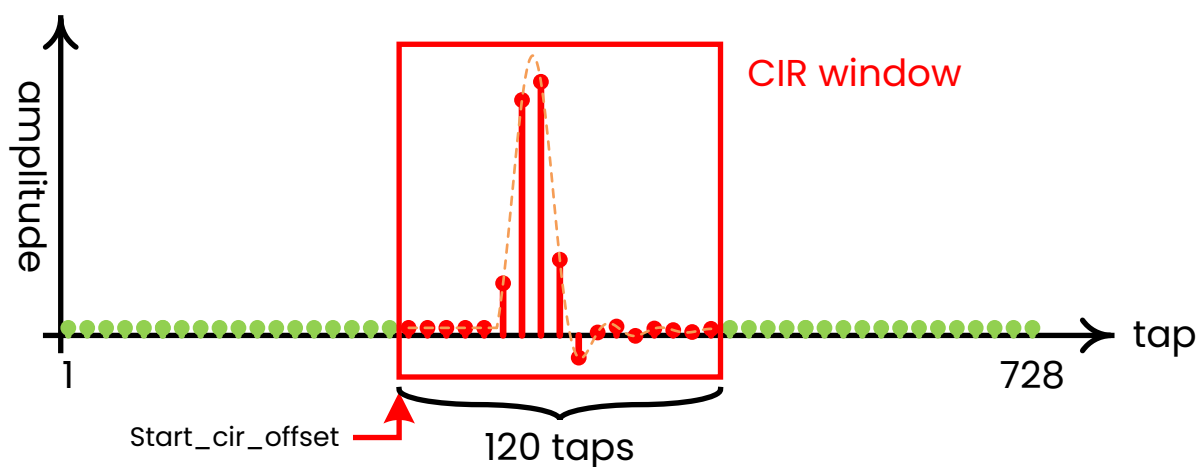


Figure 7. CIR window

To ensure that the target is captured within the 120 CIR window the starting tap of the 120 must be triggered once based on the coupled path peak (corresponding to the zero distance, as it can be considered as a direct loop back between TX and RX).

To ensure that the CIR of interest is fully captured within the 120 tap window, the data acquisition is recorded as a couple of taps earlier (before the beginning of the coupled signal).

For our reference system:

- 7 taps are considered as initial offset (In principle, this could be any number)
- It implies that the coupled path peak will appear at tap 7.

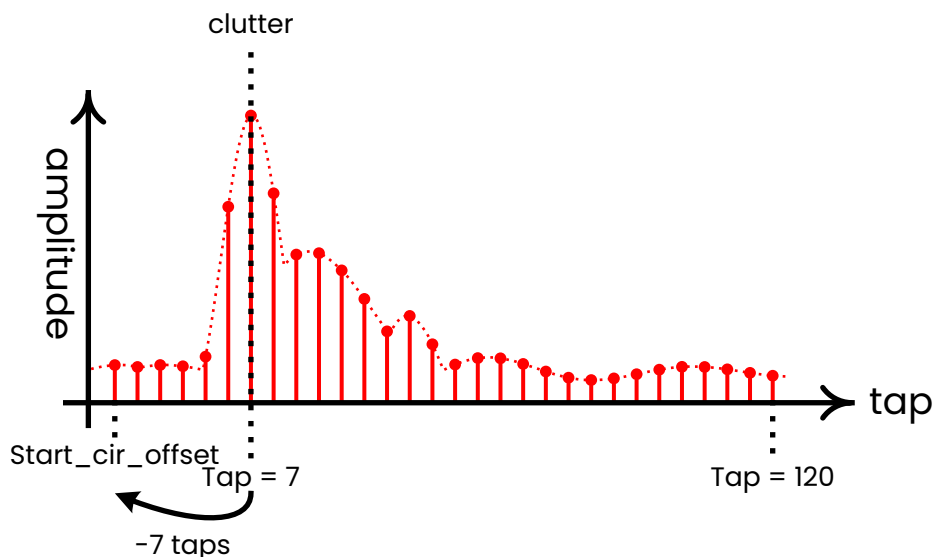


Figure 8. CIR capture window

The acquired data with sample length of 120 (or 120 taps) is composed of three parts:

1. Initial taps before the start of the coupled signal
2. Taps that capture the significant portion of the coupled signal

3. Taps that capture the tail of the coupled signal extending until the maximum practically achievable range.

The tap at which the coupled path CIR is observed corresponds to the zero-meter distance (so-called “zero distance”). It is important that this is calibrated in order to have an accurate ranging distance and avoid any distance error offset.

For more information on how to properly set the “start_cir_offset” parameter, refer to [AN14992 – SR250 calibration procedure guide](#).

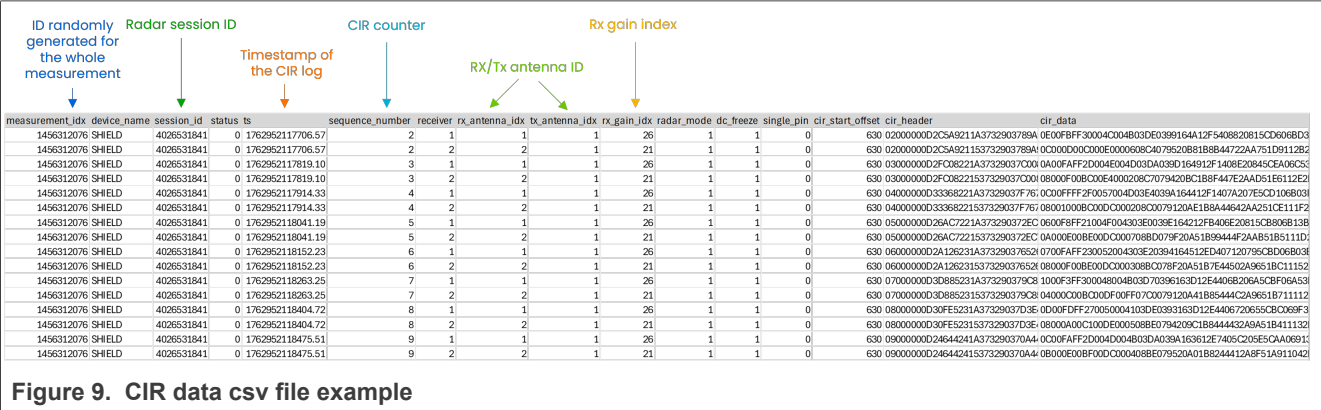
2.3 Radar RX notification

In the NXP UWB solution context, the 128 taps CIR window is sent to the host via a radar RX notification.

The UCI format of the radar RX notification is described in [UM12508 – SR250 UCI specification](#).

While running "Radar Raw Demo" provided as part of the [SR250 Plug-n-Play Application](#), CIR data is stored in a csv file for post processing.

The format of this csv file is as depicted below (one line per CIR):



Using the exported csv file, radar CIR can be plotted.

The figure below shows the plotting of raw CIR for two T/RX couples (TX1/RX1 and TX1/RX2). TX coupling can be identified at the highest peak (~tap no5). RX gain is set independently per RX chain. The value of it is defined by the AGC chain and is depending on the TX leakage amplitude.

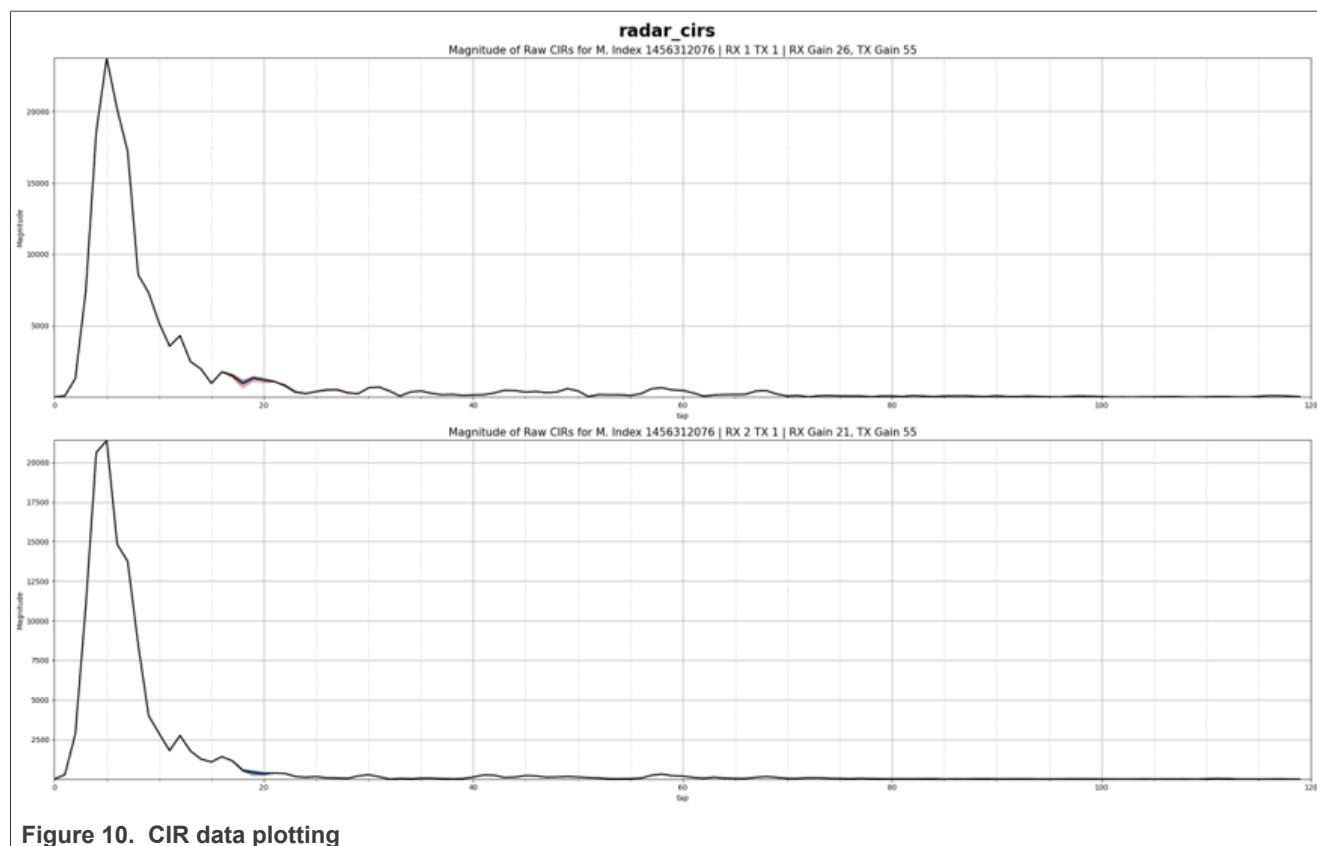


Figure 10. CIR data plotting

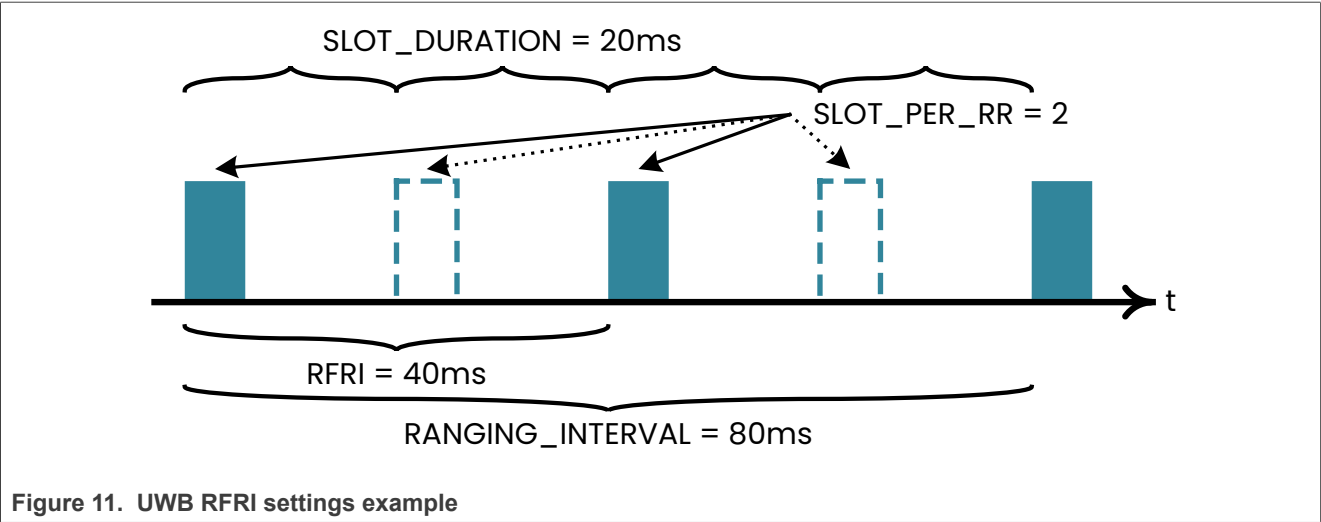
2.4 Radar Frame Repetition Interval

The radar transmitter is sending the radar frames periodically, with an interval called Radar Frame Repetition Interval (RFRI). The number of radar frames can be programmed through the **RADAR_RFRI** UCI parameter. This UCI parameter (with 0xA9 tag) is decomposed in three subparts as detailed below:

- **RANGING_INTERVAL**: same as the ranging use case it define the periodicity of the ranging (4 bytes)
- **SLOT_DURATION**: it defines a subdivision (integer) of the **RANGING_INTERVAL** (2 bytes)
- **SLOT_PER_RR**: number of active slots per **RANGING_ROUND** (1 byte)

RFRI settings for the following example:

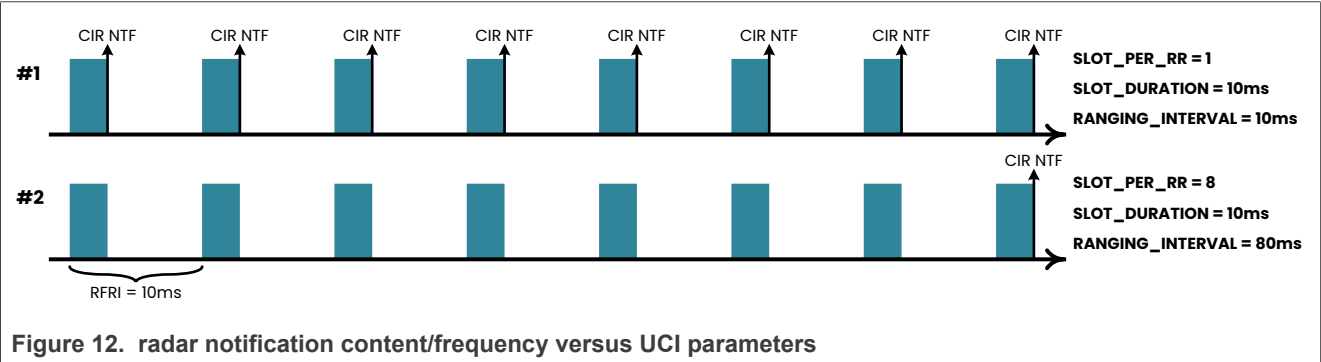
- **RANGING_INTERVAL**: defined in milliseconds ("80")
- **SLOT_DURATION**: defined in RSTU ("24000", leads to $24000/1200RSTU = 20\text{ ms}$)
- **SLOT_PER_RR**: ("2" means two frames per ranging round)



Although different UCI parameter settings can provide the same RFRI at the end, some guidelines on how to set correctly these parameters are given in the example below.

In both cases RFRI is 10 ms, so the radar coverage is as follows:

- #1 every **10 ms** UWB chip sent **1** CIR via a notification. Host is having a “live” radar picture. The drawback is that every 10 ms application processor must handle the NTF.
- #2 every **80 ms** UWB chip sent **8** CIRs via a notification. The host is having a delayed radar picture. But the advantage is that the application processor is not woken up every 10 ms.



3 Radar frame

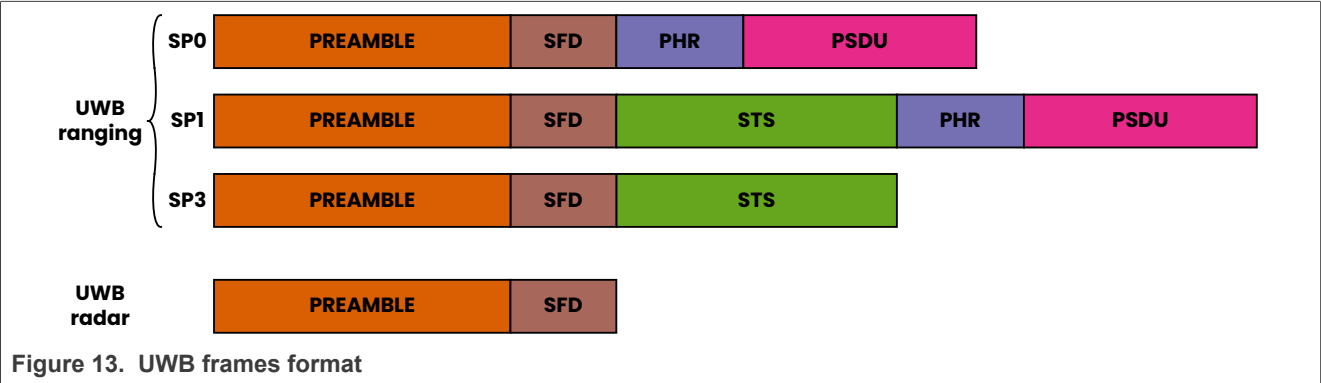
In the NXP UWB radar context:

- The IC transmitter sends a dedicated **UWB radar frame**, which consists of a specific number of pulses.
- The UWB reflected wave is then captured by the IC receiver and demodulated to extract the **CIR**.
- This CIR can be considered as one measurement of the channel at a given time t .
- This sequence is happening in a periodically giving the possibility to collect multiple CIRs. This radar period is called **Radar Frame Repetition Interval (RFRI)**.
- NXP radar post-processing is based on this batch of CIRs.

3.1 Radar frame format

For UWB radar only the preamble part of the ranging frame is used.

No STS, no PHR nor Payload are used, only the preamble/SYNC part is used.



There are currently two ways for sending UWB frames, see [Section 3.2](#) and [Section 3.3](#).

3.2 High PRF mode

3.2.1 HPRF mode - Radar frame format

In HPRF UWB frame is using the same preamble ranging structure but optimized for UWB radar. A preamble is a repetition (Nsync time) of a symbol (code Ci). In the NXP context, maximum Nsync is 256 symbols, preamble code is composed of 91 ternary code with a spreading factor of 4. A maximum of 91 x 256 pulses are sent per frame within a very short duration. For reference, this means that PRF is 111 MHz and peak PRF is 124.8 MHz. This frequency difference between mean and peak originates from only 81 of the 91 pulses are used, while 10 pulses are 0.

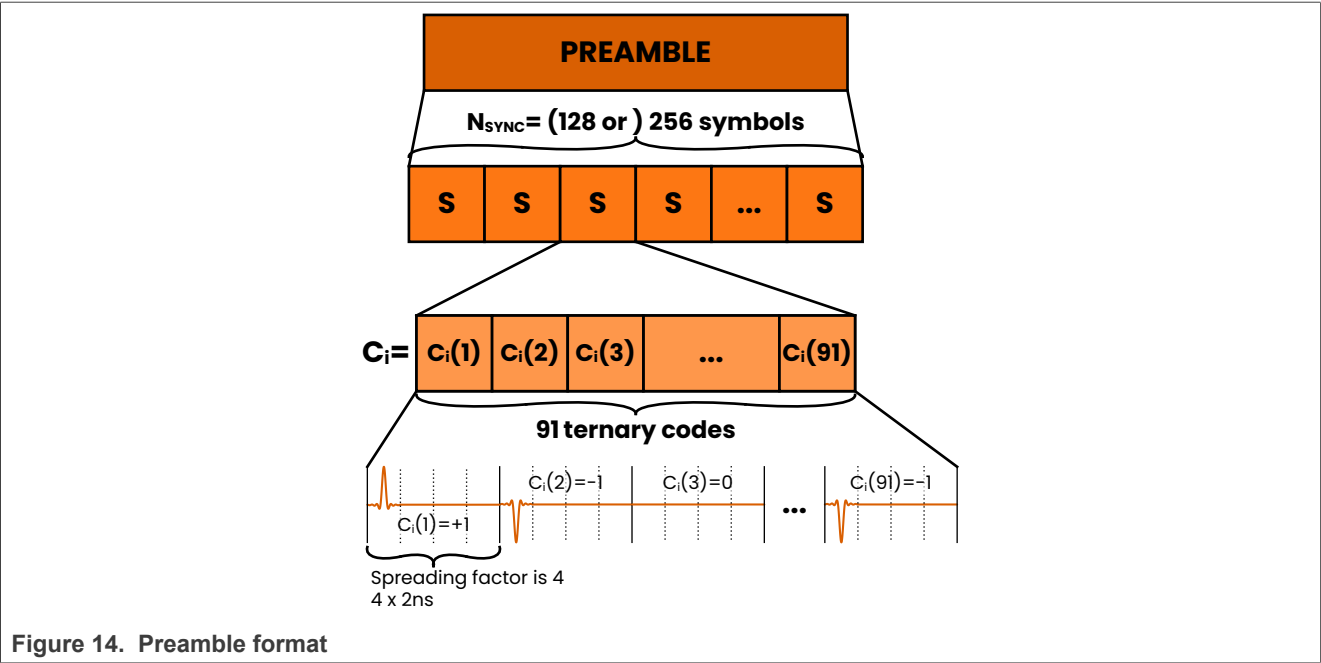


Figure 14. Preamble format

3.2.2 HPRF mode - In time domain

In HPRF mode, closely spaced pulses are sent with a code modulation that allows the receiver to perform correlation with the same code to extract the relevant information.

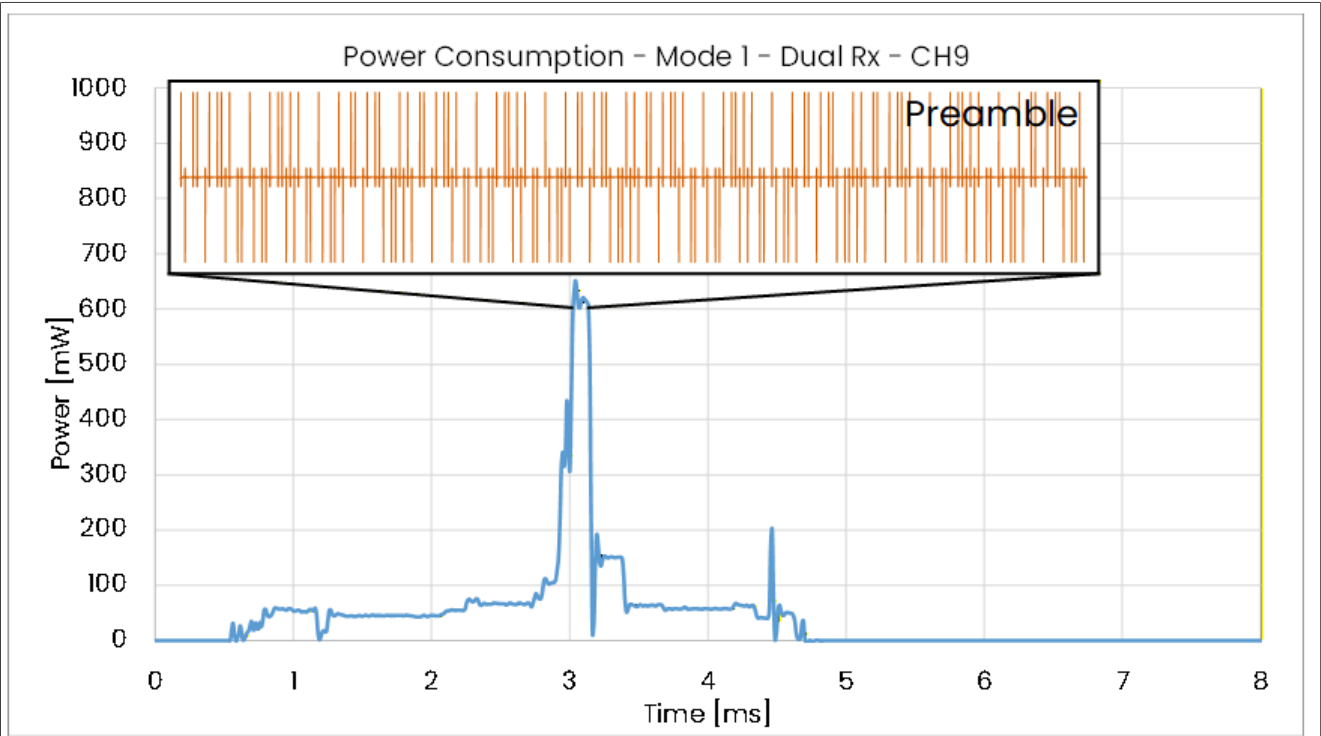


Figure 15. HPRF mode in time domain

3.2.3 HPRF mode - Demodulation

- To extract usable CIR (with sufficient amplitude and SNR), in case of HPRF, all symbols are summed up at the output of the ADC, this is what is called binning.
- This represents 91 pulses per symbol, multiplied by the number of symbol repetitions in the preamble (N_{sync} preamble symbols). At the output of the binning operation, one symbol length data is available resulting of the sum mentioned above.
- This one symbol length data is then correlated with the reference preamble code to get the CIR.

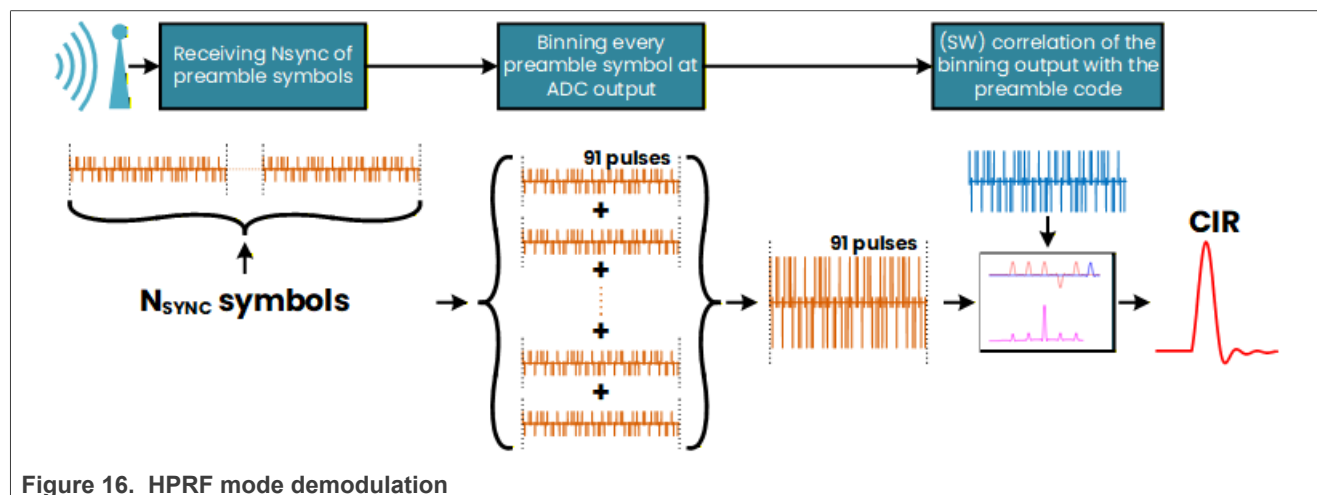
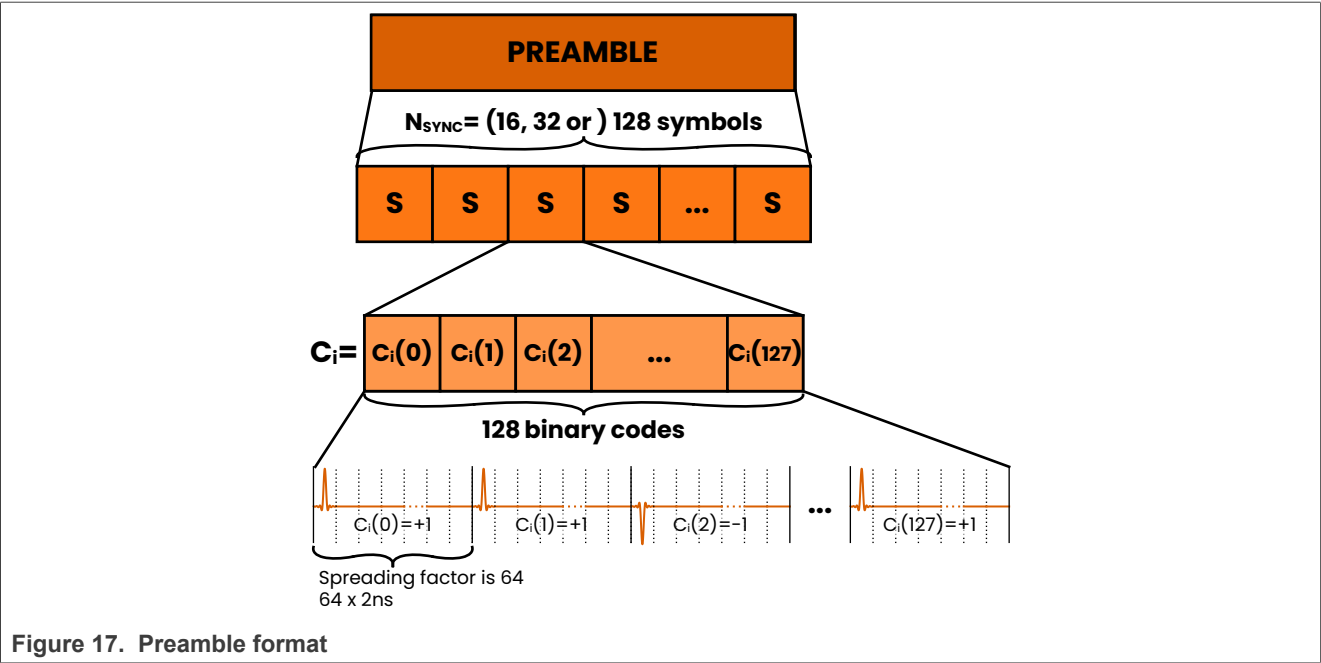


Figure 16. HPRF mode demodulation

3.3 Low PRF mode

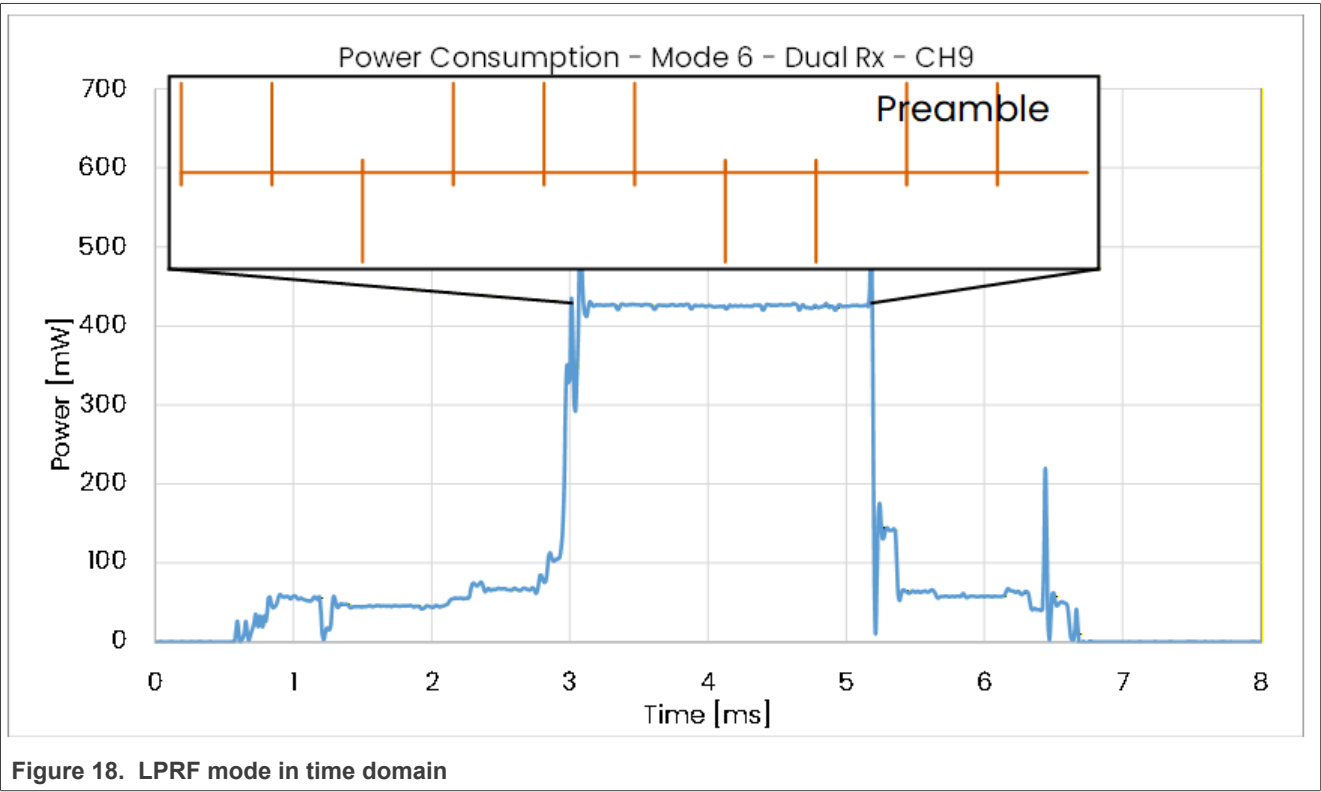
3.3.1 LPRF mode - Radar frame format

In the LPRF UWB frame is using the same preamble ranging structure but optimized for UWB radar. A preamble is a repetition (N_{sync} time) of a symbol (code C_i). In the NXP context, maximum N_{sync} is 128 symbols, preamble code is composed of 128 binary code with a spreading factor of 64. A maximum of 128×128 pulses are sent per frame within a longer short duration (compared to HPRF). The mean PRF is 7.8 MHz (the same as peak PRF)



3.3.2 LPRF mode - In time domain

In LPRF mode pulses are sent with a large pulse repetition interval (in other words, low pulse repetition frequency, PRF). Frame duration is longer, as a consequence, RX stage is also active during a longer time period compared to HPRF mode:



3.3.3 LPRF mode - Demodulation

- Similar to HPRF mode, to extract usable CIR (with sufficient amplitude and SNR), all pulses are summed up at the output of the ADC, this is what is called binning.
- This represents 127 pulses per symbol, multiplied by the number of symbol repetitions in the preamble (N_{sync} preamble symbols). At the output of the binning operation, one symbol length data is available resulting of the sum mentioned above.
- This one symbol length data is then correlated with the reference preamble code to get the CIR.

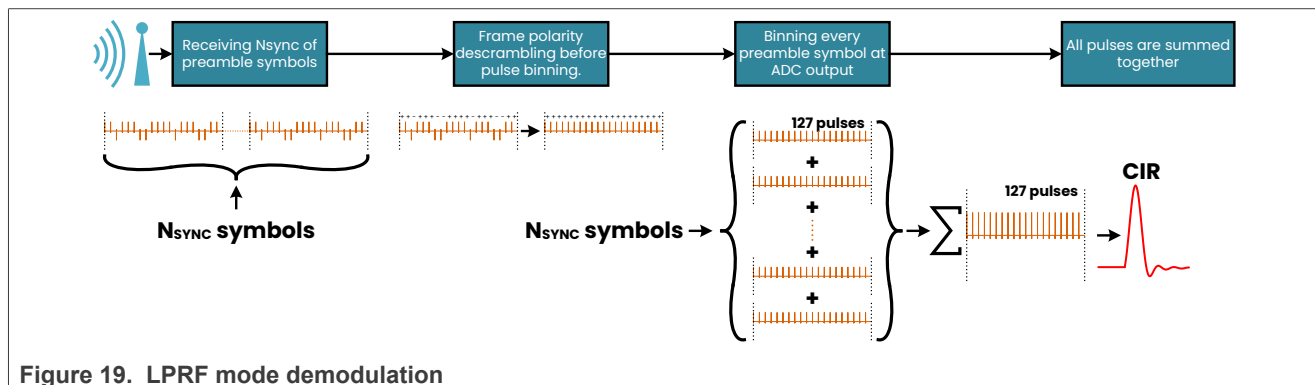


Figure 19. LPRF mode demodulation

3.4 UWB radar mode details

In total, six radar modes are available.

All these modes can be used in case of two-pin radar (that is, TX and RX are two different pins and antenna). Radar modes 4, 5, and 6 have the advantages that they can be used for single pin radar, in that case one TRX pin of the device can be used on a single antenna. However, it must be noticed that overall for single-pin radar, the AGC is set to lower gain, which has also bad NF compared to two-pin radar modes.

Table 1. UWB radar mode details

Radar mode	Type	preamble code type C_i	PRF (MHz)	Pulse (ns) (reference pulse duration)	code Length L (Nb of chips in a preamble code)	δ_L Delta Length (Nb of preamble code in a Symbol)	N_{sync} Symbol repetition (Nb of Symbol in the SYNC)	SYNC duration (μs) T_{frame}
1, 2	HPRF	typ. 26	111	2	91 (ternary)	4	128	$N_{\text{sync}} \times 0.73 \mu\text{s} = 93$
3	HPRF	typ. 26	111	2	91 (ternary)	4	256	$N_{\text{sync}} \times 0.73 \mu\text{s} = 186$
4	LPRF	typ. 95	7.8	2	127 (binary)	64	16	$N_{\text{sync}} \times 11.6 \mu\text{s} = 262$
5	LPRF	typ. 95	7.8	2	127 (binary)	64	32	$N_{\text{sync}} \times 11.6 \mu\text{s} = 524$
6	LPRF	typ. 95	7.8	2	127 (binary)	64	128	$N_{\text{sync}} \times 11.6 \mu\text{s} = 2097$

A brief description of the different modes characteristics is provided below:

- **Mode 1:** recommended for all use cases by default. This radar mode is offering a high processing and correlation gain. Detection of moving person from 15cm up to $> 10\text{m}$ depending on antenna performances. This mode is efficient from power consumption perspective as UWB RX open window is short. Static object detection is very difficult.
- **Mode 2:** can be used in case of close distance detection, that is, in case the received signal could get stronger than the coupled path signal $\Rightarrow$ reduce RX gain due to signal of interest being strong (when we start

CIR we do not have close object, only coupled path, then close object is coming to the antenna with higher amplitude => could cause saturation)

Note: *Mode 2 is usually not used.*

- **Mode 3:** for long-distance detection (more pulses => higher processing gain).
- **Mode 4** this mode allows the AGC to be to a higher value. But most of the gain in that mode comes from the processing gain (2K samples).
- **Mode 5 and 6:** More appropriate for long-range detection. These modes are similar to mode 4 but have an increased frame duration for longer range detection (500us) and longer frames of 4K samples, 16K mode 6 provides larger processing gain. Detection of moving person from 2m up to > 15m depending on antenna performances. This mode is not efficient from the power consumption perspective as UWB RX open window is long. Static object detection is possible.

4 Radar CIR post-processing

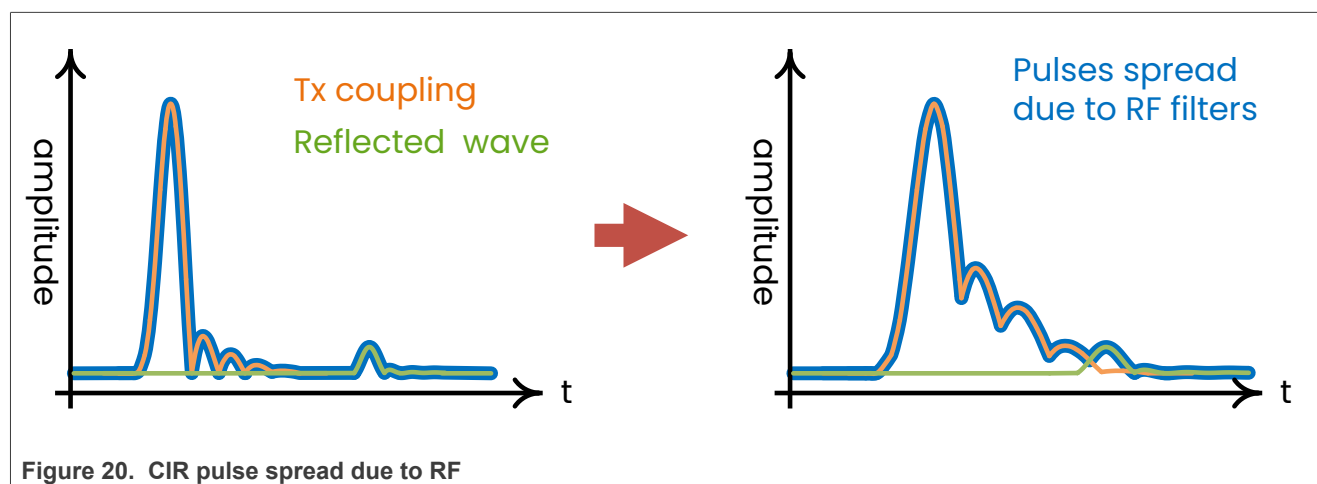
4.1 CIR tail

The narrow 2 ns pulse transmitted, the received reflected pulse and the coupled path are all going through multiple analog stages that behave as filters.

The antenna bandwidth is a limiting factor. If it does not have a sufficient frequency bandwidth, it tends to cut the UWB signal in the frequency domain and spread the pulse (in the time domain) across multiple taps. This tail effect of the different pulses (TX and RX) tends to limit the resolution of the radar system.

As a consequence, the pulses are spread in time domain, across multiple taps. Then instead of being localized in the CIR to a few “taps”, the coupled path and signal of interest are spreading across multiple taps.

This pulse spread makes it very difficult to see the signal due to reflection in the CIR, unless we find a way to **remove the coupled path**.



4.2 CIR shape depends on coupling

Depending on the platform design, the TX signal can couple via multiple paths. [Figure 21](#) shows an example of different paths via which coupling happens.

This leakage signal gets coupled to RX at multiple points:

- Within the chip:
 - The on-chip isolation is large. In the SR250 context, there is a 60 dB isolation for RX1 and 44 dB isolation for RX2.
- At the board level:
 - This depends on the board design and the layout and quality of T/R switches
- From TX to RX antenna:
 - Depends on the antenna placement. The closer the TX and RX antennas, the lower the isolation

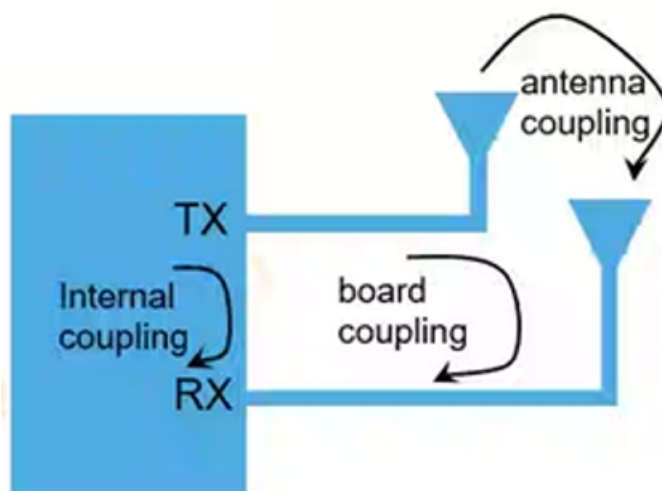


Figure 21. Coupling paths on a UWB radar device

Depending on the exact delay and attenuation of the coupled path (coupling can be constructive or destructive), the resulting CIR can have different shapes (Figure 22).

From left to right, the first picture shows an ideal CIR shape of an internal coupled path. On the picture in the middle, we can see that the CIR has more spread and less amplitude. On the last picture, the coupled path is not the main path. The CIR shape of the coupled path impacts the target detection sensitivity, especially at close distance.

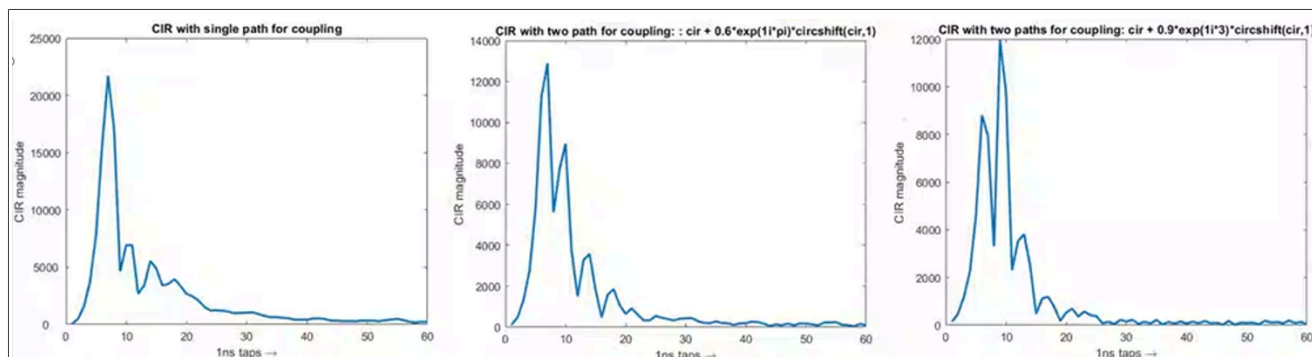


Figure 22. Coupling impacts on CIR shape

Because the clutter has a much greater amplitude than the target incoming signal, it is impossible to detect it from the tail of the clutter without removing the clutter.

4.3 TX-RX coupling path cancellation (Clutter removal) definition

TX-RX coupling path cancellation (commonly referred to as "clutter removal") removal consists in taking a reference CIR (called clutter reference) at a certain time t_0 and subtracts it to the CIRs, which are regularly captured. In some implementations, a high-pass filter (like an exponential filter) is used instead of removing one reference CIR.

After an ideal clutter removal, only the target variations remain.

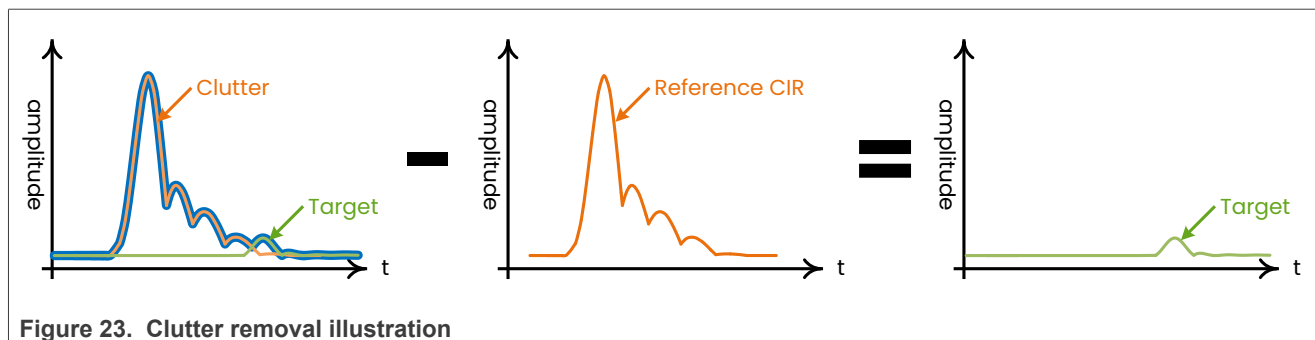


Figure 23. Clutter removal illustration

Consider a sinusoid variation of the target (due to chest movement during breathing),

the reflected waves are affected by this movement and are creating a Doppler signature around the target position in the CIR. As a reminder, the Doppler effect is affecting the frequency and phase, but not the amplitude of the signal.

Therefore, plotting the target CIR must highlight:

- A constant peak at the target position if magnitude versus distance (tap) is plotted
- A vector rotating (the center is the origin of the system) with constant amplitude, if a complex system versus time is used.

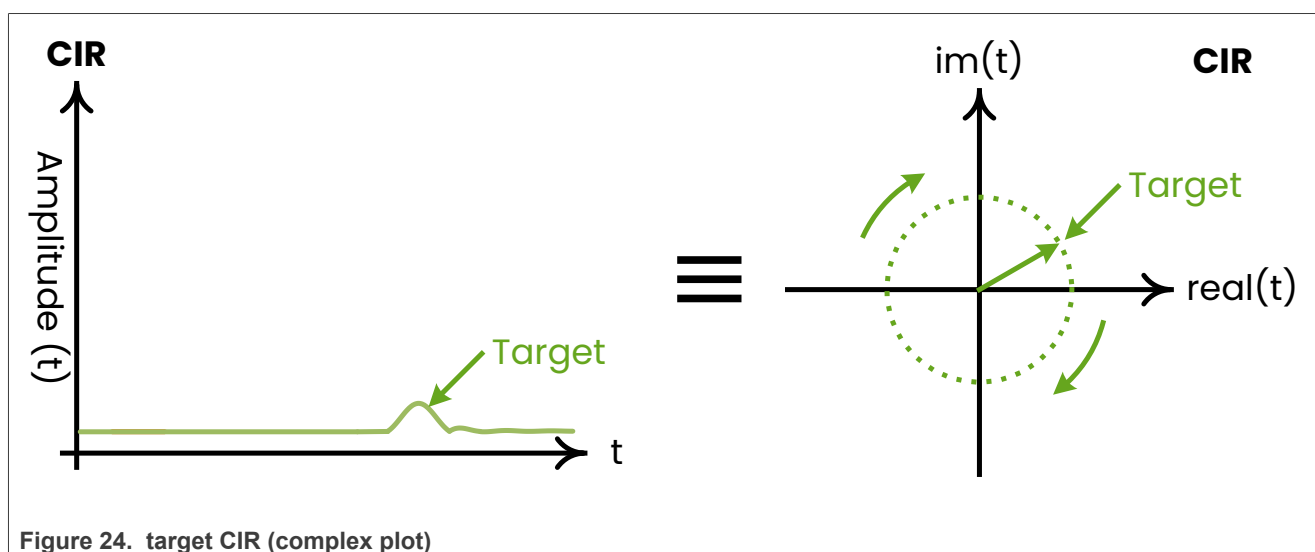
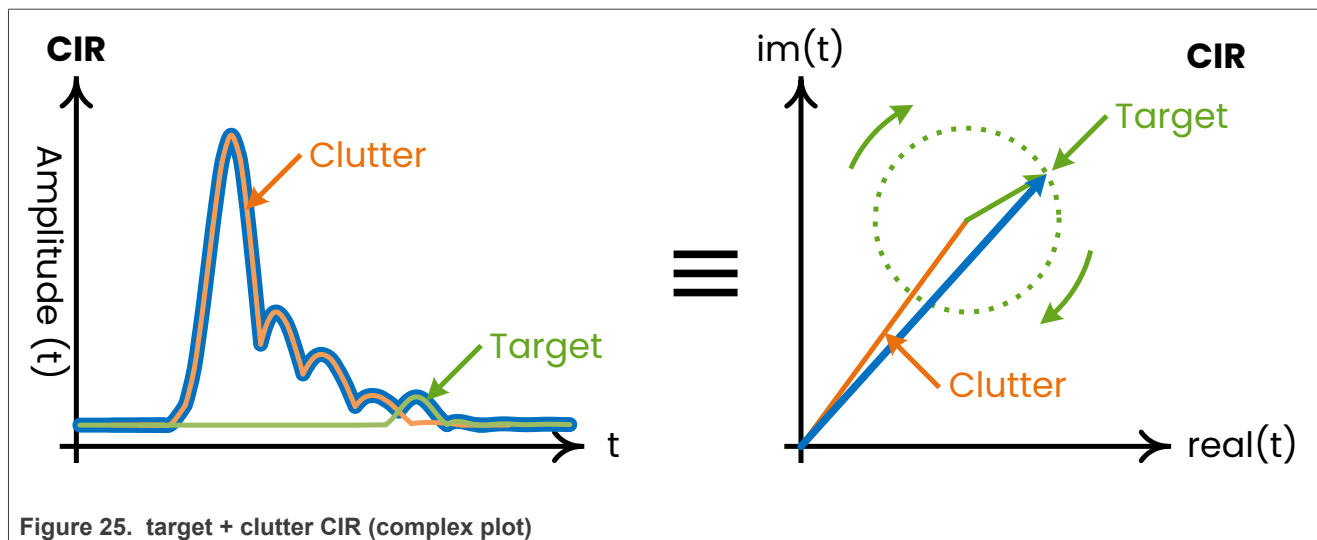


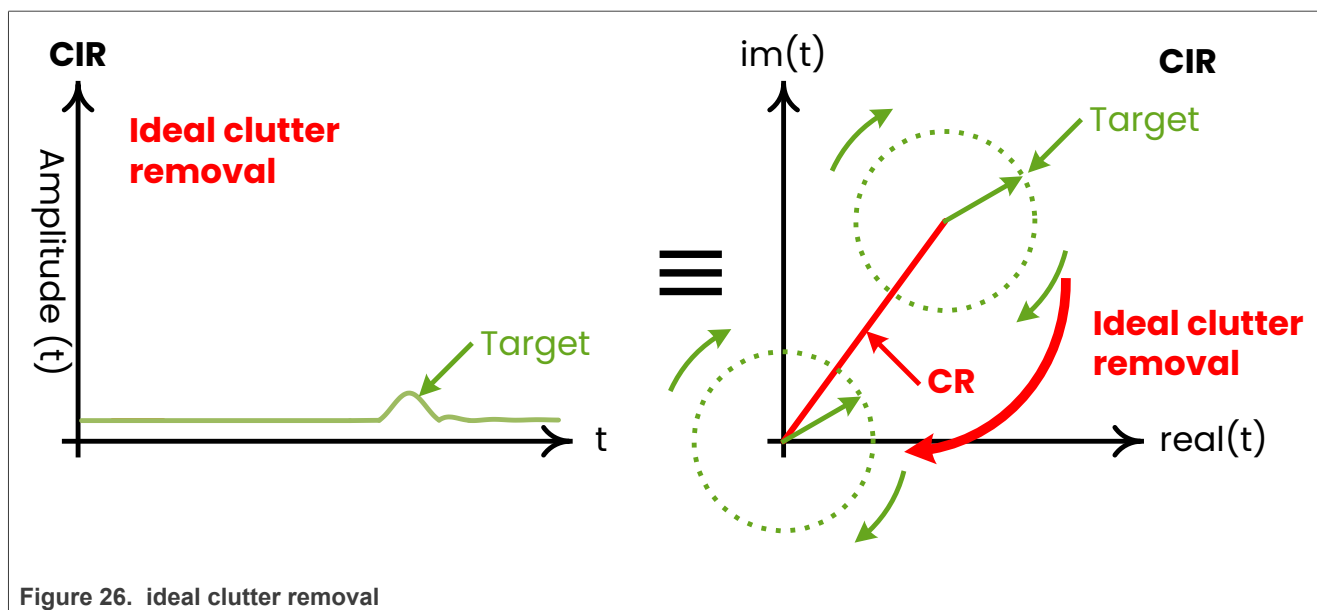
Figure 24. target CIR (complex plot)

Considering that clutter (that is, coupled path signal) does not shift overtime, CIR can be seen as the combination of a rotating vector (target) with a center rotation shifted from the complex system origin by a constant vector (clutter):



Then, in an ideal case, removing the clutter reference value to the measured CIR leads to subtracting this reference clutter vector to all CIRs.

It is like shifting back the rotating vector to the complex system origin:



4.4 Clutter reference imperfections

In previous sections, clutter removal was based on a reference clutter constant in time, which is not the case in real-life applications:

- Clutter reference (CR) is not only containing the clutter. When defined as clutter reference, the CIR is the superposition of the clutter and the target.
- After a radar frame emission/reception, IC RF and modem blocks are turned off to save power. These blocks are turned on before the next frame emission/reception. This turning on and off is impacting the IC internal temperature, which is drifting over time. This temperature drifts is also impacting on the CIR shape. Changes in CIR due to this drift can also be stronger than the changes in CIR due to the target movement and therefore

interferes with target detection. To compensate CIR temperature drift, reference CIR used for clutter removal is periodically up-dated in our reference algorithm.

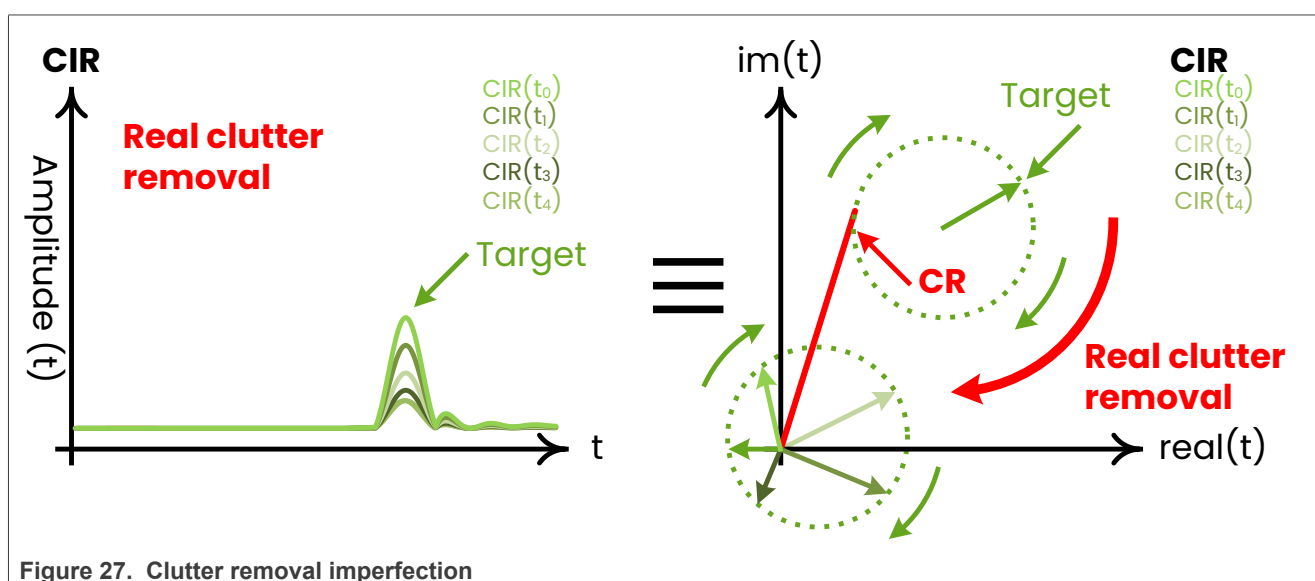
- IC blocks analog noise and imperfections also impact clutter reference.
- Also, the environment changes contribute to the clutter reference.

Due to the above phenomena, clutter removal is not ideal.

Therefore, in a complex system, the subtraction of this CR leads to a vector in rotation with a rotation center not aligned with the complex system origin.

In a Cartesian system, clutter removal leads to a residual signal composed of the target and the imperfectly canceled clutter.

In both representation target amplitude varies in time.



Note that the amplitude variation of the target CIR observed is just an artefact of the clutter removal imperfections and not related to the doppler effect.

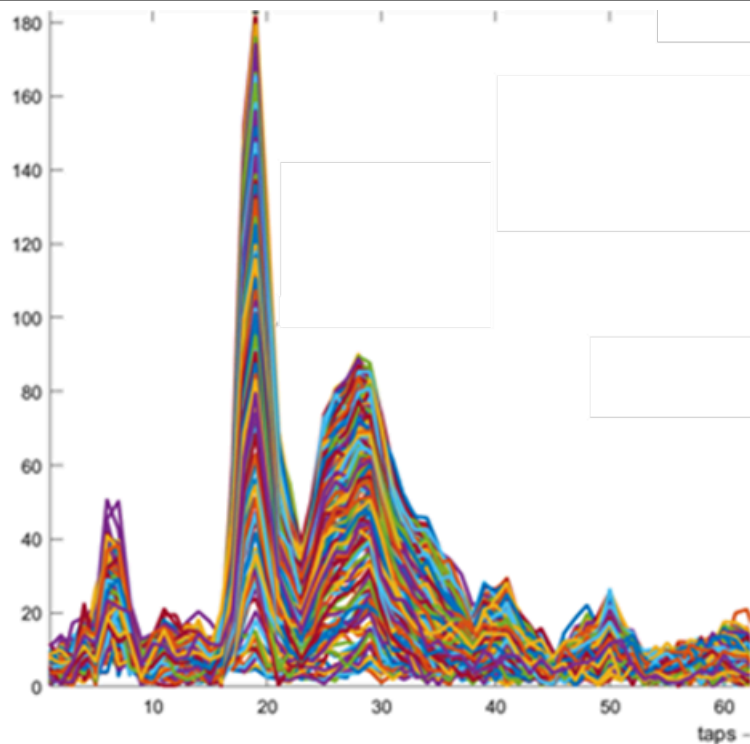


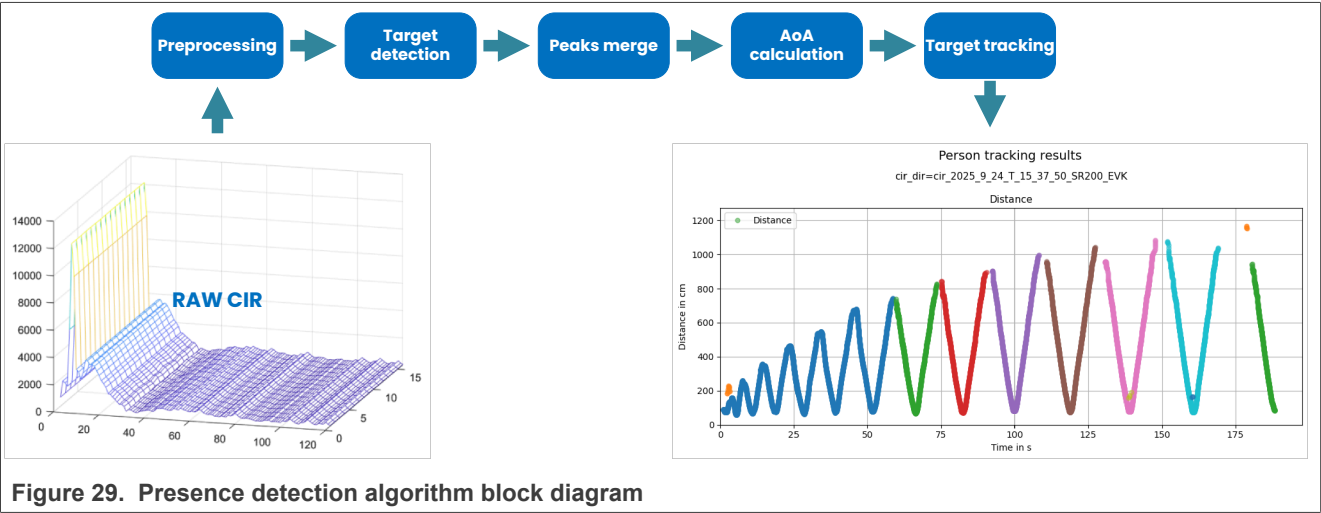
Figure 28. clutter removal imperfection plotting

5 Presence detection algorithm

Presence detection algorithm implementation could be split into 5 main stages that are running sequentially. Inputs of this presence detection algorithm are the raw radar CIRs provided by the UWB IC.

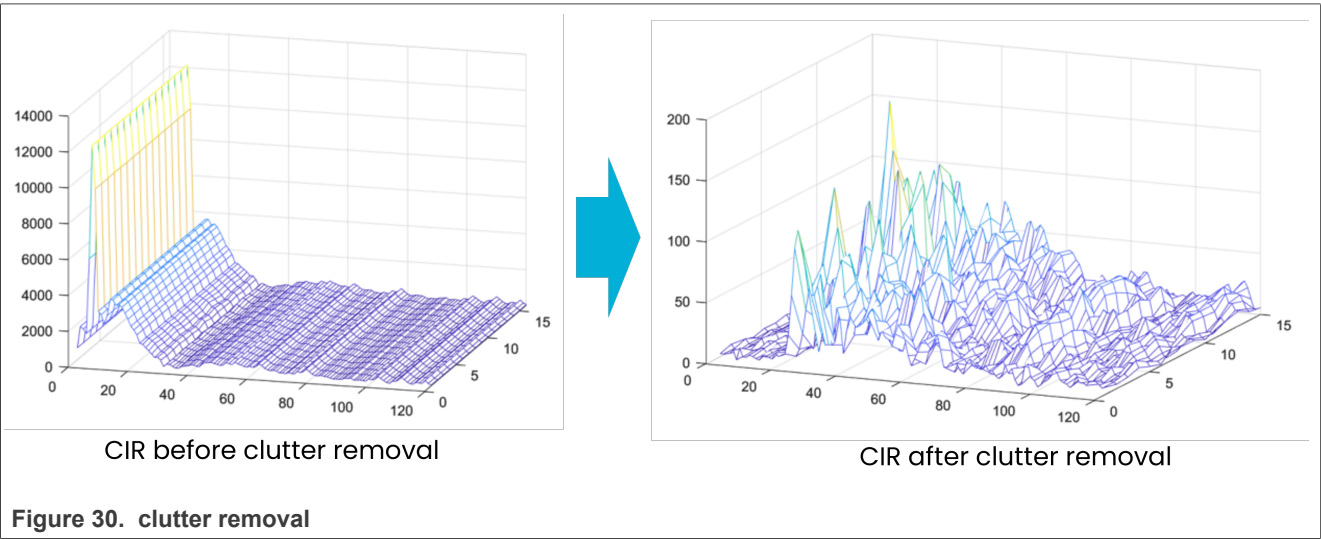
The left plot in [Figure 29](#) shows 3-axis representation of the CIR. X,Y plane shows the CIR amplitude versus tap over the 128 taps of the CIR window, highlighting the CIR position (large yellow peak). The third axis is all CIR curves over time or over each RFRI.

The right plot in [Figure 29](#) is an example of the detection of a moving target, moving back and forth from min to max distance in front of the radar device with 1m increase distance at each travel.



5.1 Pre-processing

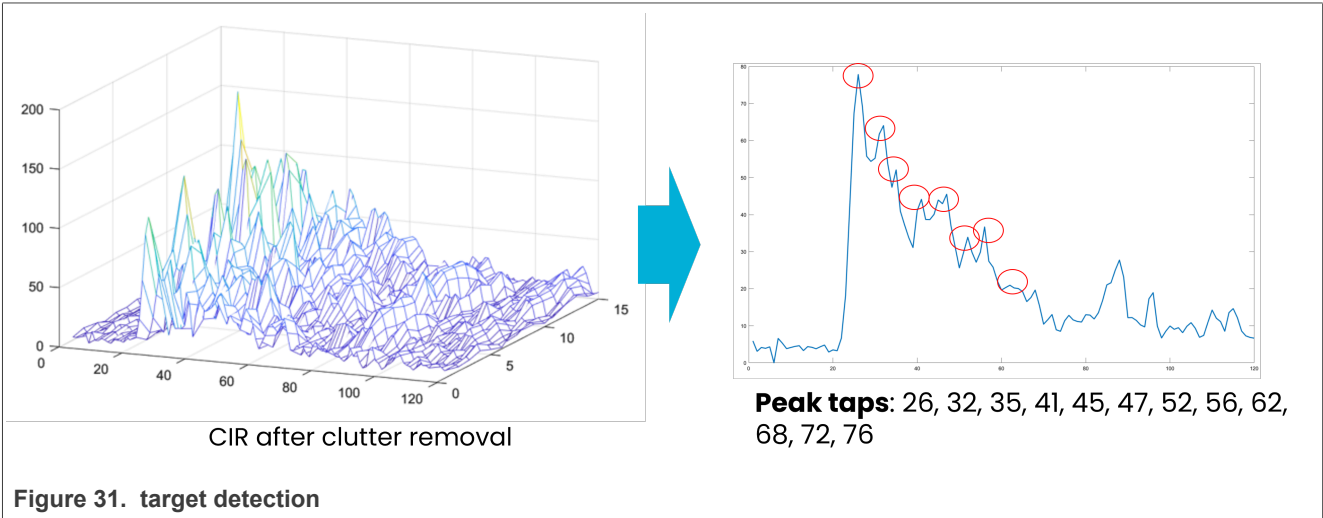
The CIRs must be pre-processed to be able to perform peak detection. Multiple CIRs are put in a CIR window and clutter-removal is performed to remove the fully static targets.



5.2 Target detection

Peak search and CFAR are then performed to detect the peaks (that is, the non-static targets). There are also multiple other steps done like Doppler frequency extraction, SNR estimation, confidence estimation.

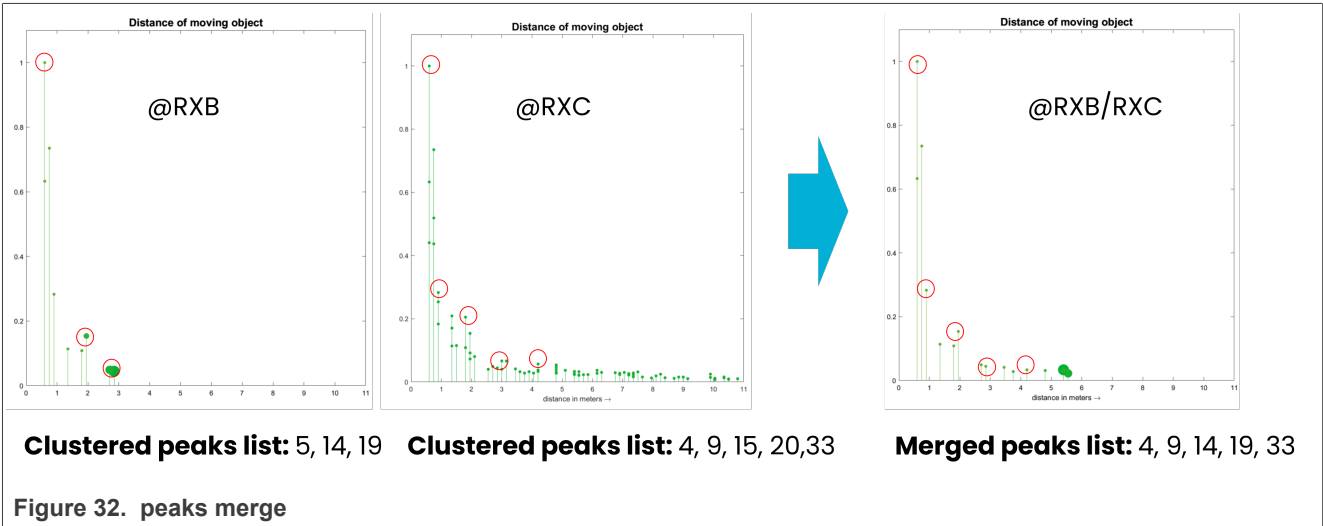
Constant false alarm rate (CFAR) detection is a common form of adaptive algorithm used in radar systems to detect target returns against a background of noise.



5.3 Merge peaks

The peak list of each RX chain is clustered (merge close peaks on the same receiver)

Then each clustered peaks list of different receivers, radar modes, and window lengths (depending on the configuration) are merged such that we have one single list of detected targets.



5.4 AoA calculation

If at least two receivers are used/available, the AoA can be calculated using the PDoA between the receivers.

Note that AoA is only computed on the defined short peak list.

In case of two people are located at the same distance from the radar (on the same circle centered on the radar device) then the reported peak will be the fusion of both target signatures. Meaning amplitude and phase will be a mixture of the two. In this case, PDoA measurement is less significant than AoA calculation (as it cannot be used to distinguish the two targets).

5.5 Target tracking

To avoid outliers and/or gaps in the detections, a Kalman filter is applied to track the targets. This results in a smoother detection computed curve with fewer outliers.

Figure below shows a target moving from a close to a far distance, and back. It can be observed that on this plot the target detection is lost at around 11m and recovered at the same distance on the return.

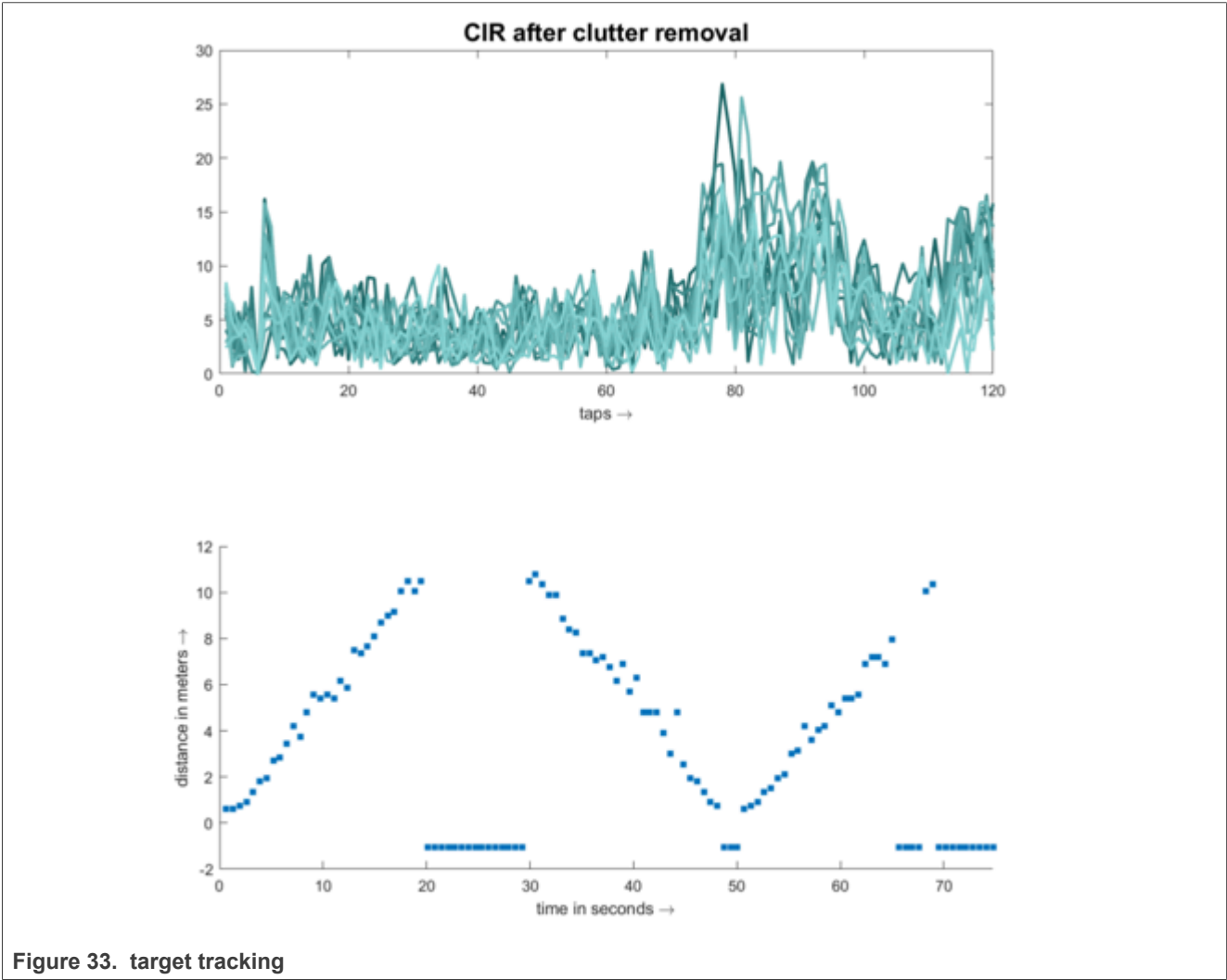


Figure 33. target tracking

6 Abbreviations

Table 2. Abbreviations

Abbreviation	Meaning
AN	Application Note
BPM	Breath Per Minute
CIR	Channel Impulse Response
FFT	Fast Fourier Transform
FSPL	Free Space Path Loss
RCS	Radar Cross Section
RFRI	Radar Frame Repetition Interval
SNR	Signal-to-Noise Ratio
UWB	Ultra-WideBand

7 References

- [1] User manual – UM12508 – SR250 UCI specification ([link](#))
- [2] Application note – AN14992 – SR250 calibration procedure guide ([link](#))

8 Revision history

Table 3. Revision history

Document ID	Release date	Description
AN13989 v.1.2	30 March 2026	Document security status changed to "public". Update for SR250.
AN13989 v.1.1	9 April 2024	Focusing on SR2XX.
AN13989 v.1.0	5 July 2023	Initial version of the document.

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Tables

Tab. 1.	UWB radar mode details	14	Tab. 3.	Revision history	27
Tab. 2.	Abbreviations	25			

Figures

Fig. 1.	A pulse radar system	2	Fig. 18.	LPRF mode in time domain	13
Fig. 2.	Radar Cross Section (RCS)	3	Fig. 19.	LPRF mode demodulation	14
Fig. 3.	Free Space Path Loss (FSPL)	3	Fig. 20.	CIR pulse spread due to RF	16
Fig. 4.	TX antenna to RX antenna coupling	4	Fig. 21.	Coupling paths on a UWB radar device	17
Fig. 5.	TX antenna to RX antenna coupled path	4	Fig. 22.	Coupling impacts on CIR shape	17
Fig. 6.	CIR window	5	Fig. 23.	Clutter removal illustration	18
Fig. 7.	CIR window	6	Fig. 24.	target CIR (complex plot)	18
Fig. 8.	CIR capture window	6	Fig. 25.	target + clutter CIR (complex plot)	19
Fig. 9.	CIR data csv file example	7	Fig. 26.	ideal clutter removal	19
Fig. 10.	CIR data plotting	8	Fig. 27.	Clutter removal imperfection	20
Fig. 11.	UWB RFRI settings example	9	Fig. 28.	clutter removal imperfection plotting	21
Fig. 12.	radar notification content/frequency versus UCI parameters	9	Fig. 29.	Presence detection algorithm block diagram	22
Fig. 13.	UWB frames format	10	Fig. 30.	clutter removal	22
Fig. 14.	Preamble format	11	Fig. 31.	target detection	23
Fig. 15.	HPRF mode in time domain	11	Fig. 32.	peaks merge	23
Fig. 16.	HPRF mode demodulation	12	Fig. 33.	target tracking	24
Fig. 17.	Preamble format	13			

Contents

1	Introduction	2
1.1	Pulse radar system	2
1.2	UWB radar sensitivity	2
1.3	Radar Cross Section (RCS)	2
1.4	Free Space Path Loss (FSPL)	3
1.5	TX-RX coupled path	3
2	Radar CIR (Channel Impulse Response)	5
2.1	CIR tap	5
2.2	CIR capture window	5
2.3	Radar RX notification	7
2.4	Radar Frame Repetition Interval	8
3	Radar frame	10
3.1	Radar frame format	10
3.2	High PRF mode	10
3.2.1	HPRF mode - Radar frame format	10
3.2.2	HPRF mode - In time domain	11
3.2.3	HPRF mode - Demodulation	12
3.3	Low PRF mode	12
3.3.1	LPRF mode - Radar frame format	12
3.3.2	LPRF mode - In time domain	13
3.3.3	LPRF mode - Demodulation	14
3.4	UWB radar mode details	14
4	Radar CIR post-processing	16
4.1	CIR tail	16
4.2	CIR shape depends on coupling	16
4.3	TX-RX coupling path cancellation (Clutter removal) definition	17
4.4	Clutter reference imperfections	19
5	Presence detection algorithm	22
5.1	Pre-processing	22
5.2	Target detection	23
5.3	Merge peaks	23
5.4	AoA calculation	23
5.5	Target tracking	24
6	Abbreviations	25
7	References	26
8	Revision history	27
	Legal information	28

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